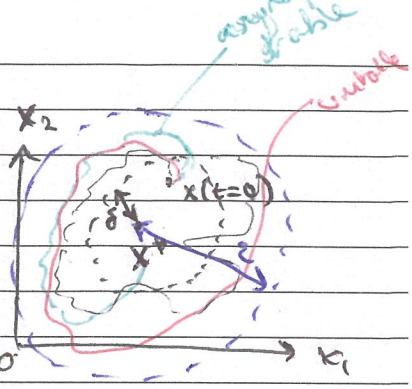
20/10/26

Mid-module Review

Linear Systems - Stability

$$\frac{d\underline{x}}{dt} = \underline{\underline{A}}\,\underline{x}$$ ($\underline{\underline{A}}$: constant matrix)



Fundamental matrix

$$\frac{d\underline{x}}{dt} = \underline{\underline{P}}(t)\,\underline{x}$$ ($\underline{\underline{P}}$: $n \times n$ matrix $= A$)

Solve → sol^n $\underline{x}^{(1)}, \underline{x}^{(2)}, \ldots, \underline{x}^{(n)}$ — $n \times n$ system

linearly independent

Fundamental matrix $\underline{\underline{\Psi}}(t) = \left(\underline{x}^{(1)} \;\; \underline{x}^{(2)} \;\; | \;\; \ldots \;\; | \;\; \underline{x}^{(n)} \right)$

Example w/p

$$\underline{\underline{P}}(t) = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$$

Solve $\frac{d\underline{x}}{dt} = \underline{\underline{P}}\,\underline{x}$

See example 4.11 for the solutions

$$\underline{x}^{(1)}(t) = \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t} \quad \text{(eigenvalue } -1\text{)}$$

(1st ↑ eigenvector)

$$\underline{x}^{(2)}(t) = \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} \quad \text{(eigenvalue } 3\text{)}$$

(2nd ↑ eigenvector)

$W[\underline{x}^{(1)}, \underline{x}^{(2)}] \neq 0$ ∴ $\underline{x}^{(1)}$ & $\underline{x}^{(2)}$ are linearly independent ∴ they form a fundamental set by def^n 4.16 and so the fundamental matrix is

$$\underline{\underline{\Psi}}(t) = \begin{pmatrix} e^{-t} & e^{3t} \\ -2e^{-t} & 2e^{3t} \end{pmatrix}$$

Gen sol^n: $\underline{x} = c_1 \underline{x}^{(1)} + c_2 \underline{x}^{(2)} + \ldots + c_n \underline{x}^{(n)}$ $\leftrightarrow$ $x_1 = c_1 x_1^{(1)} + c_2 x_1^{(2)} + \ldots + c_n x_1^{(n)}$ (4.37)

$$= \begin{pmatrix} \underline{x}^{(1)} & \underline{x}^{(2)} & \ldots & \underline{x}^{(n)} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$$

$$\underline{x}(t_0) = \underline{\underline{\Psi}}(t_0)\,\underline{c}$$

$$\underline{\underline{\Psi}}^{-1}(t_0)\,\underline{x}(t_0) = \underline{\underline{\Psi}}^{-1}(t_0)\,\underline{\underline{\Psi}}(t_0)\,\underline{c} \quad \text{on multiplying by } \underline{\underline{\Psi}}^{-1}(t_0)$$

$$= \underline{\underline{I}}\,\underline{c}$$

$$= \underline{c}$$

Gen sol^n / Wlp sol^n: $\underline{x}(t) = \underline{\underline{\Psi}}(t)\,\underline{\underline{\Psi}}^{-1}(t_0)\,\underline{x}(t_0)$ (4.39)

Generic fundamental matrix $\underline{\Psi}(t)$ – n linearly indep solutions

Special fundamental matrix $\underline{\Phi}(t)$ – " they satisfy th^m 4.20

Solution $\underline{x}(t) = \underline{\Phi}(t)\,\underline{x}(t_0)$ (4.44)

cf. $\underline{x}(t) = \underline{\Psi}(t)\,\underline{\Psi}^{-1}(t_0)\,\underline{x}(t_0)$

Example 4.21

$$\underline{P}(t) = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$$

Gen sol^n $\underline{x}(t) = c_1\begin{pmatrix} 1 \\ -2 \end{pmatrix}e^{-t} + c_2\begin{pmatrix} 1 \\ 2 \end{pmatrix}e^{3t}$

The columns of $\underline{\Phi}(t)$ are the solutions that satisfy

(i) $\underline{x}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} c_1e^0 + c_2e^0 \\ -2c_1e^0 + 2c_2e^0 \end{pmatrix} = \begin{pmatrix} c_1 + c_2 \\ -2c_1 + 2c_2 \end{pmatrix}$

Solve for c_1 and c_2 to find $c_1 = c_2 = 1/2$

$$\therefore \underline{x}^{(1)} = \begin{pmatrix} \frac{1}{2}e^{-t} + \frac{1}{2}e^{3t} \\ -\frac{1}{2}e^{-t} + \frac{1}{2}e^{3t} \end{pmatrix}$$

(ii) $\underline{x}(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} c_1 + c_2 \\ -2c_1 + 2c_2 \end{pmatrix} \Rightarrow c_1 = -1/4$ and $c_2 = 1/4$

$$\therefore \underline{x}^{(2)} = \begin{pmatrix} -\frac{1}{4}e^{-t} + \frac{1}{4}e^{3t} \\ -\frac{1}{4}e^{-t} + \frac{1}{4}e^{3t} \end{pmatrix}$$

So that the special fundamental matrix is

$$\underline{\Phi} = \begin{pmatrix} (i) & (ii) \end{pmatrix} = \begin{pmatrix} \frac{1}{2}e^{-t} + \frac{1}{2}e^{3t} & -\frac{1}{4}e^{-t} + \frac{1}{4}e^{3t} \\ -e^{-t} + e^{3t} & -\frac{1}{2}e^{-t} + \frac{1}{2}e^{3t} \end{pmatrix}$$

Nonhomogeneous Systems

$ay'' + py' + qy = g$

homog: $y'' + py' + qy = 0 \rightarrow y_c$

nonhomog: find sol^n y_p

Gen sol^n: $y_{g.s.} = y_c + y_p$.